

A Multi-Agent Path Planner for Escorted Navigation in GPS-Denied Environments

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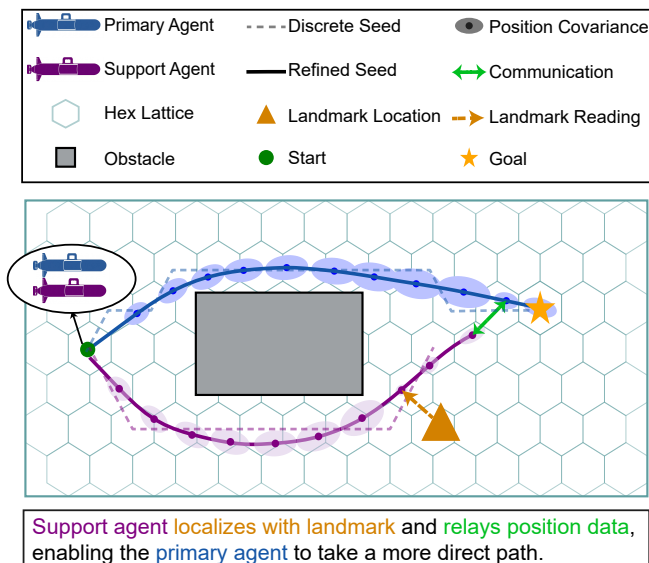


Fig. 1. The escort problem. The primary must reach the goal with its localization uncertainty under a terminal bound, and the support is routed to hold it there. Both trajectories are searched on the heading-aware hexagonal lattice and then refined as B-splines.

Abstract—Centralized multi-agent path planning with cooperative localization in GPS-denied environments is generally computationally intractable when formulated in full belief space due to tightly coupled uncertainty dynamics. Prior work either approximates belief dynamics or restricts inter-agent coupling. We study a multi-agent planning problem in which a primary agent must reach a goal in minimum time subject to a chance-constrained collision-avoidance requirement and a terminal bound on localization uncertainty. We obtain a tractable surrogate of the underlying belief-space problem by optimizing over the mean-state trajectories of all agents and propagating covariance for terminal constraint evaluation only. We propose a discrete-to-continuous pipeline combining hexagonal-grid joint search with spline-based trajectory refinement. Across 50 randomized scenarios, when the terminal uncertainty constraint is 50% of the terminal uncertainty of a reference lone primary agent plan, our planner finds solutions in 80% of the scenarios against 54% for the strongest baseline. We validate these results through the marine robotics simulator HoloOcean [1] and show that the generated paths satisfy the collision constraint in 100% and the terminal uncertainty bound in 93% of closed-loop Monte Carlo trials.

I. INTRODUCTION

In environments where GPS signal is unreliable, such as underwater, or in regions subject to spoofing or jamming, agents must rely on other forms of information for localization via GPS-denied solutions. In GPS-denied environments,

autonomous agents struggle with state uncertainty due to inertial measurement unit (IMU) drift. As the estimation error accumulates, an agent’s pose uncertainty can grow to a point where waypoint tracking degrades and mission objectives are compromised. Common planning strategies ameliorate this by biasing trajectories toward informative landmarks and introducing cooperating agents to utilize inter-agent measurements for cooperative localization [2], [3]. Consequently, trajectory planning must account for uncertainty propagation when estimation error impedes task performance. While belief-space planning provides a principled formulation for uncertainty-aware decision making, it is typically intractable in multi-agent settings due to the dimensionality of coupled belief dynamics [4], [5]. This motivates surrogate formulations that optimize mean trajectories while propagating uncertainty only for constraint evaluation.

In this work, we consider a centralized multi-agent planning problem in which a primary agent must reach a goal in minimum time while subjected to a chance-constrained collision-avoidance requirement and a terminal bound on localization uncertainty.

We compute a coordinated discrete plan over a joint state space that captures motion constraints and inter-agent ranging capabilities. This solution is then lifted into continuous space and refined into smooth, dynamically feasible trajectories that satisfy kinematic constraints.

Our contributions are as follows:

- 1) A formulation of centralized multi-agent path planning with cooperative localization (MAPP-CL) in the presence of obstacles, in which support agents are routed jointly with a primary agent subject to a terminal localization bound and chance-constrained collision avoidance under both agent and obstacle uncertainty.
- 2) A discrete-to-continuous planner, a joint search over a heading-aware hexagonal lattice followed by B-spline refinement, that enforces the curvature, collision, and terminal-uncertainty constraints in both stages.
- 3) A benchmark of 50 randomized scenarios against four baselines adapted from adjacent literature, on which the planner solves 80% of scenarios at the 50% constraint level against 54% for the strongest baseline.
- 4) Closed-loop Monte Carlo validation in HoloOcean with an independent factor-graph estimator, showing that the collision constraint holds in 100% and the terminal uncertainty bound in 93% of trials.

In the following sections, we formalize the problem, describe the discrete and continuous planning stages and

evaluate our approach in a simulated GPS-denied environment.

II. RELATED WORK

Planning under uncertainty is naturally posed as a partially observable Markov decision process, which is intractable in general [4]. Belief-space planners restrict beliefs to Gaussians and propagate them along candidate trajectories, over a roadmap [6], a tree [7], or a locally optimized path [8], usually under the assumption that the most likely observation is received [5]. These methods plan for a single agent. With several agents the joint belief and the search space grow exponentially with the size of the team.

Cooperative localization lets agents share position estimates to reduce dead-reckoning drift [2], [3] and is standard in underwater navigation [9]. Because cooperating agents share measurement history, naive fusion double counts information, and covariance intersection [10] is the usual remedy [11]. Planning with cooperative localization in the loop is more recent. Theurkauf et al. [12] grow a sampling-based tree over the joint belief of a team, and Liu et al. [13] plan risk-aware trajectories for underwater teams. Our support agents have no goals of their own and are routed only to meet the primary's terminal bound, and our joint search is exhaustive on a lattice rather than sampled.

Multi-agent path finding solves the discrete coordination problem exactly with conflict-based search [14] or approximately by planning the agents one at a time in priority order [15], [16], but it treats positions as known. Hexagonal grids reduce directional bias [17], and B-splines give smooth trajectories whose curvature can be bounded [18], [19]. Obstacle avoidance under position uncertainty follows the chance-constrained formulation of Blackmore et al. [20], and we bound each spline segment by its MINVO hull [21] so that the constraint is linear in the control points.

III. PROBLEM FORMULATION

Let $\mathcal{I} = \{1, 2, \dots, P\}$ denote a team of P agents, where agent $i = 1$ is the primary agent tasked with reaching a goal location $G \in \mathbb{R}^2$. All other agents $2, \dots, P$ are support agents whose sole task is to assist the primary agent's localization. All agents move at a common constant speed v , so minimizing time is equivalent to minimizing trajectory length, and we use arc length $s = vt$ as a shared clock throughout. The environment contains M static convex obstacles $\mathcal{O}_1, \dots, \mathcal{O}_M$, each defined as an intersection of half-planes $a_j^\top x \leq b_j$ with unit outward normals a_j and offsets b_j , and each located only up to a covariance Σ_o . The environment also contains a set $\mathcal{L}$ of landmarks with known positions $\ell \in \mathbb{R}^2$ and positional uncertainty Σ_ℓ .

A. Decision Variables

We optimize over mean-state trajectories only. The decision variables are the heading profile $\theta_i(s)$ of every agent and the primary's path length S_1 . Positions follow from the headings at the arc samples $s_k = k \Delta s$ through

$$x_i(s_{k+1}) = x_i(s_k) + \Delta s \begin{bmatrix} \cos \theta_i(s_k) \\ \sin \theta_i(s_k) \end{bmatrix}. \quad (1)$$

B. Covariance Propagation

Let $\Sigma_i(s)$ denote the 2×2 position covariance of agent i . Covariance is not a decision variable. It is propagated along the mean trajectories and used only to evaluate constraints. Between measurements the agents dead-reckon, and we model the drift as process noise whose variance grows linearly with distance and is larger along the direction of travel than across it:

$$\Sigma_i^-(s_{k+1}) = \Sigma_i(s_k) + Q(\theta_i(s_k)) \Delta s, \quad (2)$$

where $Q(\theta) = R(\theta) \text{diag}(q_{\parallel}, q_{\perp}) R(\theta)^\top$, $R(\theta)$ is the planar rotation, and $q_{\parallel} \geq q_{\perp}$ are the along-track and cross-track noise intensities.

C. Landmark Measurements

A landmark ℓ provides a range-bearing measurement whose quality decays smoothly with the distance $d_{i\ell} = \|x_i - \ell\|$ through the logistic taper

$$q_{i\ell} = \frac{1}{1 + \exp((d_{i\ell} - r_v)/w_v)}, \quad (3)$$

which is close to one inside the visibility range r_v and rolls off over a width w_v , with no hard cutoff. Every landmark contributes information weighted by its quality, and the posterior follows the information form of the Kalman update for a direct position measurement, identical to the Joseph form for $H = I$:

$$\Sigma_i(s_k) = \left(\Sigma_i^-(s_k)^{-1} + \sum_{\ell \in \mathcal{L}} q_{i\ell} (S_{i\ell} + \Sigma_\ell)^{-1} \right)^{-1}, \quad (4)$$

where $S_{i\ell} = R(\beta_{i\ell}) \text{diag}(\sigma_r^2, \sigma_b^2) R(\beta_{i\ell})^\top$ is the sensor noise rotated to the line-of-sight bearing $\beta_{i\ell}$.

D. Inter-Agent Fusion

Agents exchange position estimates at fixed arc-length intervals Λ . Their estimates share history, so adding the information directly would double count it, and we fuse with covariance intersection [10], which is consistent under unknown cross-correlation. When agents i and j communicate at separation d_{ij} , agent i receives $P_{j \rightarrow i} = (\Sigma_j + R_c)/w_{ij}$, where $R_c = \sigma_c^2 I$ is the communication noise and w_{ij} is a taper of the form (3) with half weight at the communication range r_c , and updates

$$\Sigma_i^+ = \left(\omega \Sigma_i^{-1} + (1 - \omega) P_{j \rightarrow i}^{-1} \right)^{-1}, \quad (5)$$

with $\omega \in [0, 1]$ chosen to minimize $\det \Sigma_i^+$. Agent j is updated in the same way, and as $w_{ij} \rightarrow 0$ the transferred information vanishes.

E. Optimization Problem

The planning problem collects the objective and all constraints as

$$\begin{aligned} \min_{\{\theta_i(\cdot)\}, S_1} \quad & S_1 \\ \text{s.t.} \quad & \det(\Sigma_1(S_1))^{1/4} \leq \bar{u}, \end{aligned} \quad (6a)$$

$$a_j^\top x_i(s) \geq b_j + r_a + z_\delta \sqrt{a_j^\top (\Sigma_i(s) + \Sigma_o) a_j}, \quad (6b)$$

$$|\kappa_i(s)| \leq 1/r_{\min}, \quad (6c)$$

$$x_i(0) = x_i^0, \quad \Sigma_i(0) = \Sigma^0, \quad x_1(S_1) = G. \quad (6d)$$

Terminal uncertainty (6a). The primary must arrive with bounded localization uncertainty. $\det(\Sigma)^{1/4}$ equals the standard deviation σ of an isotropic covariance $\sigma^2 I$, so $\bar{u}$ is in metres.

Obstacle avoidance (6b). Every agent must clear every obstacle with probability at least $1 - \delta$ at every instant, so for each agent i , obstacle o , and arc position s the constraint must hold for at least one face j of $\mathcal{O}_o$. The margin covers the vehicle radius r_a and the positional uncertainty of both agent and obstacle, with $z_\delta = \Phi^{-1}(1 - \delta)$ and Φ the standard normal distribution function [20]. A union bound over the PM agent-obstacle pairs bounds the instantaneous probability of any collision by $\Delta = PM\delta$.

Kinematics (6c). Each trajectory respects the minimum turn radius $r_{\min}$, where $\kappa_i = d\theta_i/ds$ is the curvature.

Boundary conditions (6d). All agents start from known positions with a common prior covariance, and the primary must end at the goal.

The belief dynamics do not appear as constraints: $\Sigma_i(s)$ is a deterministic function of the decision variables through (1)–(5), left implicit in (6). The agents are coupled only through (5), so the supports enter the problem only by shaping the primary's covariance.

IV. METHOD

Our planner solves (6) in two stages, a joint A* search over a heading-aware hexagonal lattice and a continuous B-spline refinement of the resulting plan, summarized in Algorithm 1.

A. Heading-Aware Hexagonal Graph

A corridor of hexagonal cells of width w_{hex} covers the line from start to goal [17]. A graph node pairs a cell with one of six headings spaced 60° apart, and three moves leave each node: continue straight, or turn by $\pm 60^\circ$ and step into the cell faced by the new heading. The heading changes by at most 60° per cell, which is the graph-level counterpart of (6c). Landmarks are appended as sensor nodes without edges that enter the search only through (4), a goal node absorbs all six headings of the goal cell so that the arrival heading is free, and obstacles act on beliefs during the search rather than removing cells. All-pairs shortest-path distances are precomputed on the graph and serve as the heuristic.

B. Joint Discrete Search

The first stage runs an A*-style search [22] over the joint state of the team. A search state stores, for every agent, its graph node, its covariance, and the arc distance travelled. Each expansion advances every agent by one edge, so the successors of a state are the Cartesian product of the agents' three moves. The accumulated cost g is the primary's arc distance alone, and states are expanded in order of $f = g + (1 + \varepsilon)h$, where h is the precomputed shortest-path distance from the primary's node to the goal. Supports contribute no heuristic cost, so h is admissible, and $\varepsilon > 0$ gives a weighted search that trades a factor of $(1 + \varepsilon)$ on the seed length for fewer expansions. Ties are broken toward lower primary uncertainty and then toward fewer idle supports.

During expansion each agent's covariance is propagated along its edge by (2)–(4), and (5) is applied whenever two agents reach a communication checkpoint at the same arc distance. A successor is discarded if any agent violates (6b) along its edge or if a support travels farther than the primary. Pareto pruning keeps the joint space manageable: at each joint node signature the search keeps a frontier of (g, Σ_1) labels and discards a new label if an existing one has no larger g and a covariance no larger in the positive semidefinite order. The pruning is sound because the heuristic is shared at the same node tuple and a larger covariance cannot become smaller under further propagation.

The search is anytime with respect to (6a). When a popped state places the primary at the goal and the spline seed it induces (Sec. IV-C) clears all obstacles, the state is returned if (6a) holds and otherwise kept as the incumbent if its terminal uncertainty is the lowest seen so far. If the expansion budget is exhausted, the incumbent is handed to the continuous stage, which attempts to recover feasibility. Only the terminal bound is relaxed in this way, never obstacle clearance.

Note. Six headings quantize direction to within 30° , so a lattice path that traces a smooth trajectory is at most $2/\sqrt{3}$ times longer than it. If (6) has a solution with strictly positive constraint slacks, a fine enough lattice therefore contains a feasible seed within that factor of the optimum, and the admissible search returns one no longer than it. The factor comes from the move set rather than the resolution, and removing it is the job of the refinement.

C. Continuous B-Spline Refinement

The seed is lifted into continuous space as clamped cubic B-splines, one per agent, with the seed's cell centres as control points. Support paths are padded to the primary's node count, and the primary's final control point is pinned to G . The decision vector x stacks the free control points: the interior points of the primary and all but the first point of each support, whose terminal position is free.

The constraints of (6) form a slack vector $s(x)$, feasible when $s(x) \geq 0$: the terminal bound, evaluated by propagating (2)–(5) along the sampled splines, the curvature bound, sampled from the spline derivatives, the requirement that no

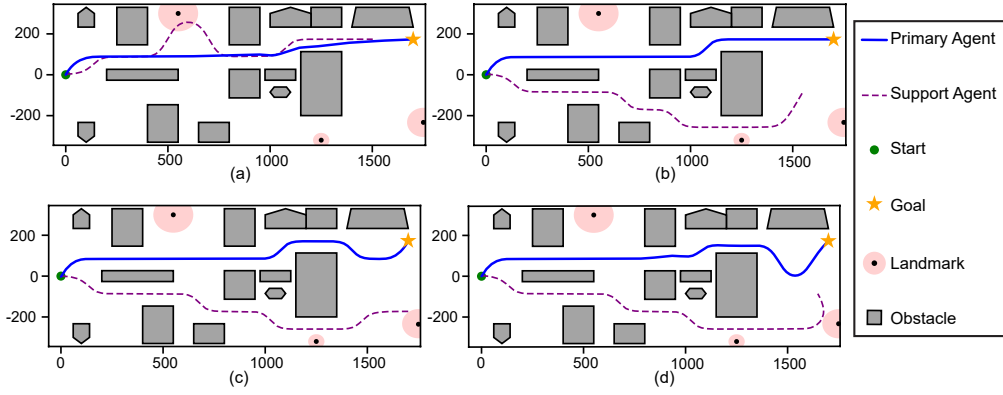


Fig. 2. Constraint ladder on one scenario: a 1700 m corridor of staggered walls, 14 obstacles and three landmarks. The terminal bound tightens from (a) to (d), and every tightening is paid for in primary length; the levels and the solutions they produce are in Table I. In (a) the support climbs into the north pocket for the weak landmark there and rejoins; in (b) it takes the southern lane to a stronger landmark and relays it from behind the last wall; in (c) the primary adds a turn before the goal so that the support can relay from closer; in (d) it dips to the centre line so that the support can reach the last landmark in time.

Algorithm 1: Discrete-to-continuous escort planning

Input : hex graph $\mathcal{G}$, start states x_i^0, Σ^0 , goal G , bound $\bar{u}$, expansion budget N
Output: B-spline control points x of every agent

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1  $h(\cdot) \leftarrow \text{ALLPAIRSSHORTESTPATHS}(\mathcal{G})$ 
2  $open \leftarrow \{\text{joint start}\}; inc \leftarrow \emptyset; seed \leftarrow \emptyset$ 
3 while  $open \neq \emptyset$  and  $expansions < N$  do
4   pop the state minimizing  $f = g + (1 + \varepsilon)h$ 
5   if primary at  $G$  and  $\text{SEEDSPINECLEAR}(\text{state})$  then
6     if  $\det(\Sigma_1)^{1/4} \leq \bar{u}$  then  $seed \leftarrow \text{state}; \text{break}$ 
7     else if  $\det(\Sigma_1)^{1/4}$  lowest so far then  $inc \leftarrow \text{state}$ 
8   end
9   foreach joint successor (every agent one edge) do
10    propagate  $\Sigma_i$  by (2)–(4); fuse by (5)
11    discard if (6b) fails or a support outruns the primary
12    insert unless dominated in  $(g, \Sigma_1)$  at its signature
13  end
14 end
15 if  $seed = \emptyset$  then  $seed \leftarrow inc$ , the least-violating candidate
16  $x \leftarrow$  control points of the seed splines;
17    $feas \leftarrow \min s(x) \geq 0$ 
18  $x^* \leftarrow x$  if  $feas$  else  $\emptyset$ ;  $x^- \leftarrow x$  // incumbents
19 foreach barrier stage; iterate until stationary or stalled do
20    $x' \leftarrow$  backtracked Adam step on  $\nabla_x J(x; \mu)$ 
21   if  $\neg feas$  then // recovery
22     accept iff  $J(x') < J(x)$ ; then  $feas \leftarrow \min s(x') \geq 0$ 
23   else accept iff  $J(x') < J(x)$  and  $\min s(x') \geq 0$ 
24   update  $x^*$  (shortest feasible) and  $x^-$  (least violating) with  $x'$ 
25   at stage end, decay  $\mu$  only once  $feas$ 
26 end
27 return  $x^*$  if  $x^* \neq \emptyset$  else  $x^-$ 

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support spline be longer than the primary's, and obstacle clearance. For the latter, each spline segment is enclosed in the convex hull of its MINVO control points [21], and each segment-obstacle pair is committed to the face of (6b) that the seed clears by the largest margin, which fixes the homotopy class and replaces the disjunction over faces with linear rows, one per hull vertex. The refinement then minimizes the primary's length under the log-barrier merit

Panel	Constraint level (% of U_{ref})	Terminal bound $\bar{u}$ (m)	Primary length (m)	Terminal σ (m)
(a)	100	13.75	1741	9.73
(b)	70	9.62	1761	8.89
(c)	50	6.87	1822	6.66
(d)	30	4.12	1890	3.91

TABLE I. The four panels of Fig. 2. The constraint level is the terminal bound as a fraction of the uncertainty an unconstrained single agent reaches on the same scenario, which is 13.75 m over a reference path of 1712 m.

function

$$J(x; \mu) = S_1(x) + \sum_k B(s_k(x); \mu), \quad (7)$$

$$B(s; \mu) = \begin{cases} -\mu \log s, & s > \delta_b, \\ -\mu (\log \delta_b + \frac{s - \delta_b}{\delta_b}), & s \leq \delta_b, \end{cases} \quad (8)$$

where the tangent extension below the knee δ_b keeps J finite for violated constraints and supplies a constant restoring gradient, so one objective serves feasible and infeasible iterates alike.

We minimize J with Adam [23] on forward-difference gradients and a backtracking line search over a schedule of barrier stages. While no feasible iterate has been accepted, as when the search handed over its incumbent, any step that decreases J is accepted and μ is held large so that the barrier drives the violation down. The first feasible iterate arms a latch: from then on only feasible steps that decrease J are accepted and μ decays between stages, shifting the effort from constraint satisfaction to shortening. The least-violating and the shortest feasible iterates are tracked throughout, and the better one is returned, so the output is never longer than a feasible seed nor more violating than an infeasible one.

V. RESULTS

A. Experimental Setup

We evaluate the planner on randomized corridor scenarios with one primary and one support. Each scenario places the goal 600 to 900 m from the start, draws two to four

landmarks with map covariances of 0.6 to 1.35 m standard deviation, placed 280 to 430 m to the side of the direct line on alternating sides so that every fix costs a detour, and draws one to three convex obstacles of up to 220 m across, centred on a triangular distribution so that most scenarios have an obstacle on the direct line. The lattice width is 150 m, the minimum turn radius 40 m, the visibility range 100 m with a 15 m taper, the communication interval 100 m, and the communication range 300 m with a 20 m taper. Dead-reckoning variance grows at $q_{||} = 0.25 \text{ m}^2$ per metre along track, a standard deviation of 5 m after 100 m, and at one ninth of that across track. The obstacle risk is $\delta = 0.05$ per agent-obstacle pair with a 5 m vehicle radius, the search runs with $\varepsilon = 0$ and 10^6 expansions, and the refinement runs 15 barrier stages sharing 1000 Adam iterations.

For every scenario we first plan a single unconstrained agent, which gives a reference length L_{ref} and terminal uncertainty U_{ref} . The bound is then tightened from 100% to 30% of U_{ref} in steps of 10%, and every planner is run at each level. A planner that fails at one level is counted as failing at the tighter ones. Scenarios in which a straight line already meets the loosest bound pose no localization problem and are discarded, which removed 34 of 84 draws to leave the 50 scenarios reported here. Lengths are reported relative to L_{ref} over the scenarios a planner solved, and wall-clock times are for whole Julia processes including start-up and graph construction.

B. Baselines

We compare against four planners adapted from adjacent literature. All share the covariance model of Sec. III, the obstacle test, and the refinement of Sec. IV-C, so the comparison isolates how the team is routed.

Greedy. The primary commits to its shortest obstacle-clear route without uncertainty reasoning, and at every step the support picks, among its three successors, the move that most reduces the primary’s uncertainty at that step, the myopic rule of greedy information gathering [24]. This removes lookahead from the support.

Formation. The primary plans alone with a weighted A* over cells and beliefs while the support holds a fixed slot in the primary’s body frame, one cell out at a 60° multiple off its heading, as in leader-follower formation control [25]. Goal candidates are accepted on the uncertainty the primary has with the support attached, and both lateral slots are tried. This removes support routing.

Sequential. Prioritized planning [15], [16]. The primary is planned first with the support parked at its start pose, under a bound slackened to twice the threshold, the support is then planned against the committed primary to minimize the primary’s terminal uncertainty, and the primary is re-planned once against the finished support path. This removes the joint state space.

CL-GBT. A port of the tree-growth strategy of the cooperative localization guided belief tree of Theurkauf et al. [12]: one rapidly exploring random tree over the joint state in continuous space, with goal-biased sampling for the primary

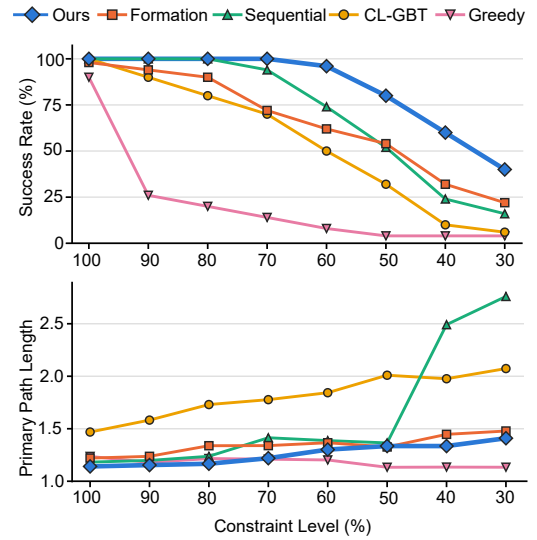


Fig. 3. Baseline comparison over the 50 randomized scenarios as the terminal uncertainty bound tightens from 100% to 30% of the single-agent reference U_{ref} . Top: fraction of scenarios solved. Bottom: mean primary path length over the solved scenarios, normalized by the single-agent reference length L_{ref} (standard deviations in Table II).

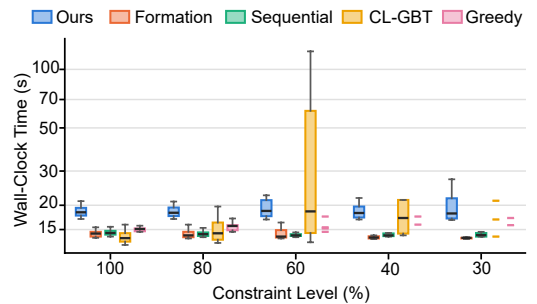


Fig. 4. Wall-clock time per planner over successful runs at five constraint levels. Boxes span the interquartile range, whiskers extend to $1.5 \times \text{IQR}$, and runs beyond the whiskers are omitted. Levels with fewer than five solved scenarios are drawn as individual runs. Only CL-GBT grows with the bound, and at 60% it leaves the axis: that box is cut at the top, with its upper quartile at 61 s and its whisker at 124 s.

and a bias that snaps the support’s sample onto the primary’s. Its belief and measurement models are replaced by ours and its iteration budget is set for equal compute time. This replaces exhaustive lattice search with sampling.

C. Comparison With Baselines

Fig. 3 reports success rate and path length against the constraint level, and Table II lists the lengths with their spread. Every planner but Greedy solves nearly every scenario at the loosest level, and the planners separate as the bound tightens. Ours solves every scenario down to 70%, 96% at 60%, 80% at 50%, and 40% at 30%. Sequential matches it down to 80% but falls to 52% at 50% and 16% at 30%, and its paths grow to more than twice the reference length, since at a tight bound the primary must detour to landmarks on its own before the support is planned. Formation degrades more gracefully and is the strongest baseline from 50% down, with 54% at half the reference bound and 22% at 30%. CL-GBT solves half the scenarios at 60% and 6% at 30%, with paths 1.5 to 2

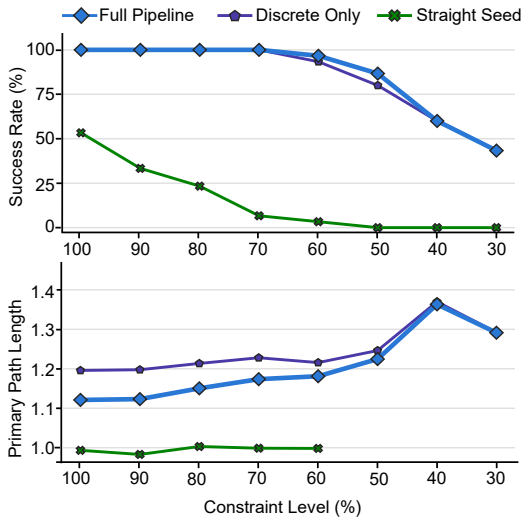


Fig. 5. Ablation on a separate set of 30 scenarios drawn without the straight-line screen. Same axes as Fig. 3: top, success rate; bottom, mean path length over the solved scenarios (standard deviations in Table II).

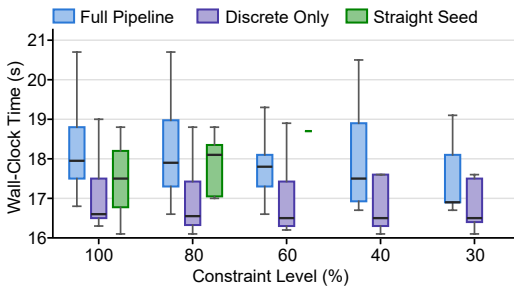


Fig. 6. Wall-clock time of the ablation arms over successful runs, drawn as in Fig. 4 on a linear axis. The refinement adds about 1.5 s to the discrete stage. The straight seed’s single run at 60% is drawn as a tick.

times the reference length because a random tree returns the first path it finds. Greedy solves 90% at the loosest level and almost nothing below it, since a support without lookahead rarely reaches a landmark in time.

The price of the higher success rate is modest. Our paths are the shortest among the solved scenarios at every level, from $1.14 L_{\text{ref}}$ at 100% to $1.41 L_{\text{ref}}$ at 30%, with the smallest spread across scenarios. Fig. 4 reports wall-clock time: a median of 18 s per run against 14 s for Formation and Sequential and 16 s for the single-agent reference, most of it shared overhead. Only CL-GBT grows with the bound, and its longest run took 730 s.

D. Ablation

The ablation asks what each stage contributes. It runs on a separate set of 30 scenarios drawn without the straight-line screen, since one arm is the straight seed itself, and compares the full pipeline with the discrete stage alone and with the refinement seeded from a straight line (Fig. 5 and the bottom block of Table II). The discrete stage alone solves the same scenarios down to 70% and one or two fewer at 60% and 50%, with paths 2 to 7% longer at every level down to 50%, which is the length the refinement recovers by cutting the

Method	Constraint level (% of U_{ref})			
	100	70	50	30
Ours	1.14 ± 0.09	1.22 ± 0.23	1.34 ± 0.26	1.41 ± 0.21
Formation	1.22 ± 0.12	1.34 ± 0.25	1.33 ± 0.16	1.48 ± 0.45
Sequential	1.18 ± 0.14	1.41 ± 0.92	1.36 ± 0.50	2.76 ± 1.51
CL-GBT	1.47 ± 0.33	1.78 ± 0.43	2.01 ± 0.41	2.07 ± 0.26
Greedy	1.24 ± 0.10	1.21 ± 0.06	1.13 ± 0.01	1.13 ± 0.01
Full pipeline	1.12 ± 0.14	1.17 ± 0.23	1.22 ± 0.22	1.29 ± 0.22
Discrete only	1.20 ± 0.15	1.23 ± 0.23	1.25 ± 0.22	1.29 ± 0.22
Straight seed + cont.	0.99 ± 0.04	1.00 ± 0.00	—	—

TABLE II. Mean $\pm$ SD of the primary path length over the solved scenarios, normalized by L_{ref} , at four constraint levels. Top: the 50 screened scenarios of Figs. 3–4. Bottom: the 30-scenario ablation set of Figs. 5–6. “—”: no scenario solved.

lattice corners. The straight seed solves 53% of the scenarios at the loosest level and none below 60%. Its solutions are not all straight lines. Of the 16 at the loosest level, 11 started inside an obstacle by 7 to 55 m, and the barrier deflected them clear at a cost of under 4% of the straight length in all but one. In seven of those the deflected line is 7 to 44% shorter than the route of the full pipeline, which went around the same obstacle by whole cells: the direct passage exists in continuous space but not on the lattice (Sec. VII). Below 80% the straight seed fails because recovery moves the path by metres, and reaching a landmark takes a detour of hundreds. The joint search finds the landmarks, and the refinement makes the route short, at a cost of about 1.5 s (Fig. 6).

VI. SIMULATION VALIDATION

Every planner in Sec. V propagates the same covariance model, so the benchmark cannot tell whether that model is calibrated. We therefore fly a planned trajectory with a vehicle model, a controller, and a state estimator that share none of the planner’s equations, and check whether the certified constraints hold over repeated runs with different noise seeds.

A. Simulator and Vehicle

We use HoloOcean [1]. Each agent is a torpedo-shaped autonomous underwater vehicle with six-degree-of-freedom Fossen dynamics [26], driven by the simulator’s depth and heading autopilot at 30 Hz at about 2.2 m/s. An outer-loop waypoint follower tracks the planned B-spline, resampled every 25 m, with a 15 m arrival radius set by the turn radius, and it steers on the estimated position, dead-reckoned forward between estimator updates, so estimation error feeds back into the executed path. The planar field is placed on one depth plane with the agents staggered in depth, so only agent-obstacle collisions are under test. Three cadences are scheduled on arc length accumulated from ground truth: the controller runs every tick, the estimator adds a pose node every 5 m, and an acoustic communication is attempted every 100 m and succeeds only within the planner’s communication range. Measurements are ground truth with additive Gaussian noise, so the simulation tests the consistency of the belief model rather than the realism of any sensor.

B. Factor-Graph State Estimator

The estimator is an existing GTSAM factor-graph stack [27], [28], developed independently of the planner; what we supply is its configuration and, through Table III, its noise. It is an incremental smoother [27] over $SE(3)$ pose nodes, one per 5 m of arc for each agent, returning the trajectory $\hat{\mathcal{X}}$ that minimizes the weighted residuals r_i of the factors $\mathcal{F}$ the graph holds,

$$\hat{\mathcal{X}} = \arg \min_{\mathcal{X}} \sum_{i \in \mathcal{F}} \|r_i(\mathcal{X})\|_{\Sigma_i}^2. \quad (9)$$

The graph is built in two layers. Each agent carries a dead-reckoning graph of its own, holding a between-factor for the odometry of each node, a unary depth factor, a unary three-axis orientation factor, and a range and bearing factor to every landmark inside the visibility range. Landmarks are not free variables: each is pinned by a prior carrying the planner’s map covariance, so the map is known and the problem is localization rather than mapping. This graph is what the agent steers on, and for a support it is also the source of the pose and the covariance its acoustic packets carry. The primary holds a second graph, and the covariance compared against the planner’s is read from that one. It takes the primary’s own nodes as they are created, and a support’s nodes in a batch when a communication succeeds, seeded with the pose that support solved for itself, together with one inter-agent factor per communication. Measurements enter both layers once rather than being drawn again, and every one of them is ground truth plus additive Gaussian noise: odometry is synthesized from the simulator’s pose sensor, and there is no inertial or Doppler sensor model.

Three configuration choices decide what the comparison in Sec. VI-C measures, and all three were in force for the runs reported there. Poses are estimated directly in the world frame: the start pose is known from the plan, so no separate map-origin node is estimated, and the covariance compared against the planner’s is the xy block of the world-pose marginal. A successful communication delivers the sender’s buffered nodes into the primary’s graph together with a relative-pose factor between the two vehicles, whose translation noise carries the planner’s own link weight; a bearing-only factor would transfer almost no position information at these separations, and the primary could never inherit the support’s landmark fix. Finally, every factor’s noise is derived from the planner’s model rather than set independently, so that the two codebases measure uncertainty with one ruler.

C. Monte Carlo Results

We flew the behind-wall plan of Fig. 8 30 times with different noise seeds. The scenario hides its one landmark behind a wall 400 m long, so only the support can see it, under a terminal bound of 1.8 m against which the planner predicts 1.80 m. Every run reached the goal within 2% of the planned length and none penetrated the wall, against an allocated collision probability of $\Delta = 0.10$ (Table IV). The estimator’s terminal marginal was under the bound in 28 of 30 runs, with the two exceptions at 1.85 and 1.96 m.

Factor	Variables	σ
Odometry (between, per 5 m node)	$\mathbf{T}_{k-1}, \mathbf{T}_k$	$1^\circ/\text{axis}; 0.22, 0.02, 0.01 \text{ m}$
Depth (unary)	$\mathbf{T}_k$	0.5 m
Compass (unary)	$\mathbf{T}_k$	$0.86^\circ/\text{axis}$
Landmark az/el/range	$\mathbf{T}_k, \ell$	$1.05^\circ, 1.05^\circ, 2.09 \text{ m}; r < 100 \text{ m}$
Inter-agent relative pose	$\mathbf{T}_k^s, \mathbf{T}_k^p$	$0.86^\circ/\text{axis}; 0.5/\sqrt{w} \text{ m}$
Initial pose prior	$\mathbf{T}_0$	$10^\circ/\text{axis}; 1.15, 0.76, 0.5 \text{ m}$

TABLE III. Factor noise of the simulated estimator, derived from the planner’s noise model so that both codebases share one ruler: along-track odometry from $q_{\parallel}$, the compass from $q_{\perp}$, landmark noise from the planner’s per-pass value rescaled to one sighting per 5 m node, and communication noise σ_c divided by $\sqrt{w}$ for link weight w . Poses are estimated in the world frame; the prior on the known start pose carries the planner’s Σ^0 in x, y . Depth is the one channel the estimator treats more conservatively than the simulation: its factor keeps the stack’s own 0.5 m against the 0.01 m actually injected.



Fig. 7. The two vehicles in HoloOcean [1]. Each is a torpedo-shaped autonomous underwater vehicle flying its planned spline at a fixed depth over the kelp seabed of the simulator’s open-water world.

The sample covariance of the true terminal error gives 1.70 m, within 6% of the prediction, while the estimator’s mean marginal of 1.51 m is more optimistic than both. The true error also carried a mean offset of 2.6 m along track and 0.8 m across, a bias of the estimate that a covariance comparison does not capture. The 2σ cross-track envelope in Fig. 8 has a half-width of about 4 m along most of the 1.1 km route and 8 m on the primary’s final bend, with a median cross-track error of 2.5 m RMS. The baselines are not re-flown, because they share the belief model whose calibration is the quantity under test.

VII. DISCUSSION AND LIMITATIONS

The discrete stage is complete only with respect to the lattice. The lattice is laid over the corridor without regard to the obstacles, so the lattice paths that clear them are neither a subset nor a superset of the continuous feasible set, and a route that exists only between two cells is not found. The refinement can shorten a seed and recover a small violation but cannot change its homotopy class. The ablation shows the cost: on seven of its 30 scenarios the refinement, started from a straight line, deflected the primary a few metres around an obstacle and finished 7 to 44% shorter than the lattice route, which had detoured around the same obstacle by whole cells (Sec. V-D). A finer lattice narrows the gap at the cost of a larger graph, and the all-pairs heuristic is cubic in the number of nodes.

The belief model is Gaussian and assumes that the planned trajectory receives its expected measurements [5]. The simulation tests calibration against an independent estimator fed

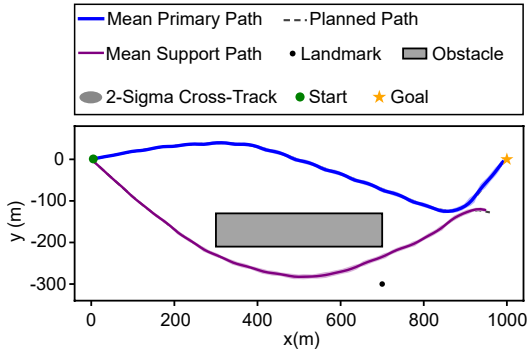


Fig. 8. Closed-loop Monte Carlo of the behind-wall plan in HoloOcean, 30 runs with different noise seeds: planned trajectories (dashed) against the mean executed trajectory of each agent, with the $\pm 2\sigma$ cross-track envelope. The primary passes north of the wall while the support diverts south to the landmark and rejoins before the goal.

Case	Runs	Collision rate ($\leq \Delta$)	Terminal bound met	Pred./emp. σ_T
Behind wall	30	0/30	28/30	1.80/1.70

TABLE IV. Monte Carlo constraint satisfaction in HoloOcean for the behind-wall plan of Fig. 8 (two agents, one obstacle, one landmark, $\bar{u} = 1.8$ m), 30 runs. Collision rate counts runs with any obstacle penetration against the allocated $\Delta = 0.10$; the terminal bound is met when the estimator's marginal at the goal is under $\bar{u}$; σ_T is the planner's prediction against the sample covariance of the true terminal error.

ground-truth measurements plus Gaussian noise, so Table IV says the planner's covariance is consistent with that estimator under the noise of Table III, not that either models an acoustic modem or a sonar. The along-track bias in the terminal error shows that a consistent covariance does not rule out a biased estimate.

The plan is centralized, computed offline, and executed open loop on a communication schedule fixed in arc length; replanning after a missed fix is not addressed. The joint search grows exponentially in the number of agents, and all experiments use one support. In the benchmark it cost 1.3 to 1.9 times the wall clock of the single-agent reference on average, most of it fixed overhead, but the exponent will bind for larger teams. Finally, the environment is planar with static convex obstacles of known position covariance, and the vehicle is modelled by a turn-radius bound alone.

VIII. CONCLUSION

We presented a centralized planner for a primary agent that must reach a goal in minimum time in a GPS-denied environment while support agents are routed to keep its localization uncertainty within a terminal bound. Optimizing over mean trajectories and propagating covariance only for constraint evaluation reduces the belief-space problem to a joint lattice search followed by B-spline refinement, with chance-constrained obstacle avoidance and a curvature bound enforced in both stages. On 50 randomized scenarios the planner solved 80% of the scenarios at half the reference uncertainty against 54% for the strongest baseline, and closed-loop Monte Carlo trials in HoloOcean showed that the certified constraints hold on the executed trajectories. Future

work includes receding-horizon replanning, scaling the joint search to larger teams, and hardware validation.

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